Use-Case Flow Specification: Request a Ride

Use Case Information

- **Use Case ID:** UC-009
- **Use Case Name:** Request a Ride
- **Primary Actor:** Rider Commuter
- **Supporting Actor:** Driver Host
- **Goal:** To enable a Rider Commuter to search for overlapping commute routes, verify 1.5 km detour compatibility, view estimated fuel cost share breakdowns, and submit a seat booking request to a Driver Host for approval.

Preconditions

1. Both Rider Commuter and Driver Host are registered, authenticated, and logged into the system.
2. The Driver Host has a verified profile with valid driving credentials and a published daily commute route with available seats.
3. The Rider Commuter has set an active commute origin, destination, and departure time window.

Postconditions

1. The ride booking status is updated to **"Confirmed"** in the system repository.
2. The available seat count for the Driver Host's scheduled trip is decremented by one.
3. An **estimated** fuel cost share split is computed and attached to the pending booking request (with the final itemized fuel-cost share generated upon completion of the shared ride per `FR-002`).
4. Real-time booking confirmations and itinerary notifications are dispatched to both the Rider Commuter and Driver Host.

Main Success Scenario

1. **Search Carpool Rides:** The Rider Commuter initiates a search for carpool matches by providing their daily pickup/dropoff locations and departure time window.
2. **Calculate Route Compatibility (`<<include>>` UC-007):** The system automatically evaluates candidate Driver Host routes against the Rider Commuter's path to determine if the added detour distance is within the mandatory **1.5 km detour tolerance threshold** (`FR-001`).
3. **Calculate Fuel Cost Share (`<<include>>` UC-011):** Calculate the estimated fuel cost share for the compatible route based on estimated trip distance, vehicle fuel efficiency, and current fuel prices. This

estimate is displayed to the Rider Commuter during the ride-request process. *(Note: Per requirement `FR-002`, the final itemized fuel-cost share is generated upon completion of the shared ride).*

4. **Display Matches:** The system displays a sorted list of compatible carpool matches showing driver ratings, estimated pickup time, added detour distance, and estimated fuel cost share.
5. **Submit Request:** The Rider Commuter selects a preferred Driver Host route and submits a seat request.
6. **Set Pending State:** The system places the seat in a temporary hold state ("Pending Approval") and sends an instant booking notification to the Driver Host (`FR-005`).
7. **Driver Approval:** The Driver Host reviews the passenger profile, pickup point, and detour metric, and accepts the ride request within the 30-minute response window.
8. **Confirm Booking:** The system updates the booking status to **"Confirmed"**, updates vehicle seat availability, and sends final itinerary details to both actors.

Alternate Flows

Alternate Flow 3a: Detour Tolerance Exceeded

- **Trigger:** The calculated added detour distance for a candidate Driver Host route exceeds **1.5 km**.
- **Flow:**
- The system flags the route as incompatible under requirement `FR-001`.
- The system excludes the route from the search results.
- If no routes fall within 1.5 km, the system displays: "*No compatible carpool matches found within your 1.5 km detour preference.*"
- The Rider Commuter is prompted to adjust their departure time window or pickup location.

Alternate Flow 7a: Driver Host Rejects Request or Request Expires

- **Trigger:** The Driver Host manually declines the request or fails to respond within the 30-minute window.
- **Flow:**
- The system updates the booking status to **"Declined"** or **"Expired"**.
- The temporary seat hold is released back to the Driver Host's available capacity.
- The system sends a notification to the Rider Commuter informing them of the refusal.
- The system prompts the Rider Commuter to select another match from their search results list.